Exercise 1

– Query to retrieve All columns

1. SELECT *
FROM Employees;

Id	First_Name	Last_Name	Department	Salary	hire_date	City
1	John	Doe	IT	55000	2018/06/15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Fransico
7	Robert	Wilson	Finance	59000	2017-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San fransico
6	Emily	Davis	IT	62000	2015-02-14	Chicago

2. – Finding Unique departments

SELECT DISTINCT (department)
FROM Employees;

Department
IT
HR
Finance
Marketing

ORDER BY fxn

3. SELECT FirstName,
LastName,
Salary
FROM Employees
ORDER BY Salary DESC;

FirstName	LastName	Salary
Emily	Davis	62 000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Daniel	Clark	53 000
Sarah	Brown	53 000
David	White	52 000
Jessica	Moore	51 000
Laura	Hall	50 000
Jane	Smith	48 000

Q:4

~ Top highest Employees

```
SELECT FirstName,
       LastName,
       Salary
FROM employees
ORDER BY Salary DESC
LIMIT 5;
```

FirstName	LastName	Salary
Emily	Davis	62000
Mike	Johnson	60 000
Robert	Wilson	59 000
John	Doe	55 000
Sarah	Brow	53 000
Daniel	Clarke	53 000

Q.5

WHERE STATEMENT,

employees working in IT.

5)

```
SELECT  First_Name,
        Last_Name,
        Department
FROM    employees
WHERE   employee = 'IT';
```

First_Name	Last_Name	Department
John	Doe	IT
Sarah	Brown	IT
Emily	Davis	IT
Laura	Hall	IT

Q.6

AND Statement

6. employees in Finance department with a salary >58000

```
SELECT  FirstName,
        LastName,
        Department,
        Salary
FROM    employees
WHERE   department = 'Finance' AND salary > 58000;
```

First_Name	LastName	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000

7. Employees in HR or Marketing

```
SELECT  First_Name,
        Last_Name,
        Department
FROM    Employees
WHERE   department 'HR OR 'Marketing;'
```

First_Name	Last_Name	Department
Jane	Smith	HR
David	White	Marketing
Jessica	Moore	HR
Daniel	Clarke	Marketing

NOT statement

8. SELECT First_Name,
Last_Name,
department
FROM Employees
WHERE department NOT IN ('HR');

First_Name	Last_Name	Department
John	Doe	IT
Mike	Johnson	Finance
Sarah	Brown	IT
David	White	Marketing
Emily	Davis	IT
Robert	Wilson	Finance
Daniel	Clark	Marketing
Laura	Hall	IT

IN

IN HR, IT & Finance

9. SELECT First_Name,
Last_Name,
Department
FROM Employees
WHERE department IN ('HR', 'IT', 'Finance');

First_Name	Last_Name	Department
John	Doe	IT
Jane	Smith	HR
Mike	Johnson	Finance
Sarah	Brown	IT
Emily	Davis	IT
Robert	Wilson	Finance
Jessica	Moore	HR
Laura	Hall	IT

Combine Conditions:

Find employees in IT with salary >50000 and are in New York.

```
SELECT  First_Name,
        Last_Name,
        Department,
        Salary,
        City
FROM employees
WHERE   salary >50000 AND department = 'IT' AND
        City = 'New-York';
```

First_Name	Last_Name	Department	Salary	City
John	Doe	IT	55000	New York
Sarah	Brown	IT	53000	New York

WHERE, AND ~~and~~ ORDER BY

11.
```
SELECT FirstName,
       LastName,
       Department,
       Salary
FROM Employees
WHERE department IN ('marketing', 'Finance') AND
      salary > 52 000
ORDER BY Salary DESC;
```

FirstName	LastName	Department	Salary
Mike	Johnson	Finance	60 000
Robert	Wilson	Finance	59 000
Daniel	Clark	Marketing	53 000

Q.12 R11

Unique cities of work excluding those in HR & IT.

```
SELECT DISTINCT (city),
       First_Name,
       Last_Name,
       department
FROM Employees
WHERE department NOT IN ('HR', 'IT');
```

First_Name	Last_Name	Department	City
Mike	Johnson	Finance	Los Angeles
David	White	Marketing	San Francisco
Robert	Wilson	Finance	Houston
Daniel	Clark	Marketing	Chicago

Q.13 Employees not in finance with salary > 50 000 ordered by ascending hire date.

```
SELECT First_Name,
       Last_Name,
       Department,
       Salary,
       hire_date
FROM Employees
WHERE department NOT IN ('Finance') AND SALARY > 50000
ORDER BY hire_date ASC;
```

Q1.3 Table Output

First_Name	Last_Name	Department	Salary	hire_date
Emily	Davis	IT	62000	2015-02-14
Paul	White	Marketing	52000	2016-04-10
John	Doe	IT	53000	2018-06-15
Sarah	Brown	IT	53000	2021-03-25
Daniel	Clark	Marketing	53000	2022-06-01

Q.14

```
SELECT  First_Name,
        Last_Name,
        Department,
        City
FROM    Employees
WHERE   City
        IN ('Los Angeles', 'Chicago') AND
        department IN ('IT', 'Marketing').
LIMIT 3;
```

First_Name	Last_Name	Department	City
Emily	Davis	IT	Chicago
Daniel	Clark	Marketing	Chicago